A PROJECT REPORT

On

ECO-WANDER FOR SUSTAINABLE TRAVEL

Submitted in partial fulfillment of the requirement of
University of Mumbai for

Web Programming Mini Project
In
Computer Engineering (IV SEM)

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PROJECT APPROVAL FOR

This project entitled "Eco-wander for sustainable travel " by Neha Zanje, Anushka Sinha, Tanishqa Temgire, and Chaitra Unnithan are approved for the degree of Bachelor of Engineering in Computer Engineering.

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DEPARTMENT OF COMPUTER ENGINEERING

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DECLARATION

We declare that this written submission for the Web Programming Mini Project entitled “Eco-Wanderer for sustainable travel” represents our ideas in our own words and where others' ideas or words have been included. We have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any ideas / data / fact / source in our submission. We understand that any violation of the above will cause disciplinary action by the institute and also evoke penal action from the sources which have not been properly cited or from whom prior permission have not been taken when needed.

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Abstract

The Eco Wonder application is an innovative AI-powered platform designed to promote environmental awareness and sustainable living through interactive digital solutions. It integrates artificial intelligence with eco-conscious practices to engage users in learning about environmental conservation in a creative and accessible way. The app allows users to generate eco-themed content, explore sustainability ideas, and receive personalized suggestions for adopting environmentally friendly habits. By transforming simple user inputs into meaningful outputs such as visuals, tips, and awareness content, Eco Wonder makes complex environmental concepts easier to understand and more engaging.

Furthermore, the application aims to bridge the gap between technology and environmental responsibility by encouraging individuals to make greener choices in their daily lives. It also fosters community participation by providing a platform where users can share ideas and spread awareness about important issues like climate change and sustainability. Overall, Eco Wonder represents a powerful combination of artificial intelligence and environmental education, helping to inspire users toward building a more sustainable and eco-friendly future.

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Chapter 1
Introduction to Eco Wonder

1.1 Introduction

In recent years, the rapid advancement of Artificial Intelligence (AI) has transformed the way users interact with digital applications. AI-powered platforms are now capable of generating creative content, analyzing user inputs, and delivering personalized experiences in real time. The Eco Wonder application is designed as an innovative solution that combines AI technology with environmental awareness, offering users an interactive and engaging platform to explore ecological concepts.

Eco Wonder allows users to input ideas, prompts, or queries related to nature and sustainability, and generates meaningful outputs such as visual representations, insights, or suggestions. This makes complex environmental topics easier to understand while encouraging creativity and participation. The application aims to bridge the gap between technology and environmental education by making learning more engaging and accessible.

Chapter 2

Objective

The main objectives of the Eco Wonder application are:

- To promote environmental awareness through interactive AI-generated content
- To provide a creative platform for users to explore ecological ideas
- To make environmental education more engaging and accessible
- To encourage users to think about sustainable solutions
- To utilize AI for generating real-time, personalized outputs

2.1 Problem Definition

Traditional methods of teaching and learning about environmental issues often lack engagement and interactivity. Many users, especially students, find it difficult to relate to theoretical concepts presented in textbooks or static content.

Additionally, there is a lack of platforms that combine:

- Environmental education
- Creativity
- Artificial intelligence

This results in reduced awareness and limited interest in sustainability topics. The Eco Wonder application addresses this problem by providing a dynamic and interactive platform where users can learn, create, and explore environmental concepts using AI.

2.2. System Requirements

2.2.1 Hardware Requirements

- Processor: Intel i3 or higher (i5 recommended)
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 500 MB free space
- Internet connection (for AI processing and cloud features)
- Any modern smartphone or computer device

2.2.2 Software Requirements

- Operating System: Windows 10/11 / Android / iOS

- Web Browser: Google Chrome / Microsoft Edge (latest version)
- Backend Support: Cloud-based AI services (e.g., AI APIs)
- Frontend: Web or mobile-based UI framework
- Database (if used): Firebase / Supabase / SQL-based system

2.2.3 Functional Requirements

- User login/signup (optional if included in app)
- Input prompt system for users
- AI-based content generation (text/image/eco insights)
- Display of generated results in real-time
- Option to refresh or regenerate outputs
- Simple navigation interface

2.2.4 Non-Functional Requirements

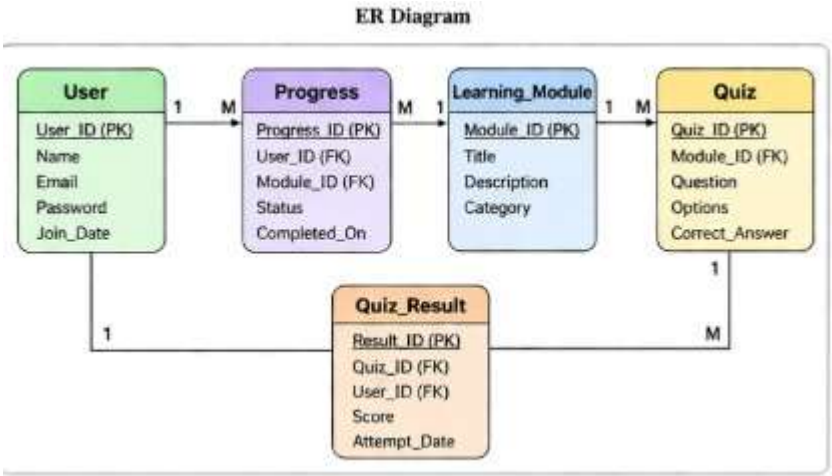
- Fast response time for AI outputs
- Scalable system to support multiple users
- Secure handling of user data
- User-friendly and responsive UI design
- High availability and reliability
- Cross-platform compatibility (mobile + web)

Chapter 3
Methodology

The database of the Eco Wonder application is designed to store user data, eco-friendly activities, and AI-generated suggestions. The main entities include User, Activity, Suggestions, and Feedback. Each user can perform multiple activities and receive multiple eco-suggestions. The database ensures efficient storage and retrieval of environmental data.

An Entity-Relationship (ER) diagram represents entities and their relationships in a system and is widely used in database design to visualize structured data.

3.1 Database schema and ER diagram



3.2 Algorithms or Technologies Used and their description

Eco Wonder uses AI-based content generation and interactive learning methods to deliver environmental education.

Technologies Used:

- Artificial Intelligence (AI) for content generation
- Web/Mobile development framework (React / Flutter)
- Cloud database (Firebase / Supabase)
- JSON-based data handling

3.2.1 Algorithm / Tools / Techniques

1. Content Recommendation Algorithm
    - Suggests environmental topics based on user activity
    - Tracks user learning history and recommends next modules
2. Quiz Evaluation Algorithm

- Compares user answers with correct answers
- Generates score and feedback instantly

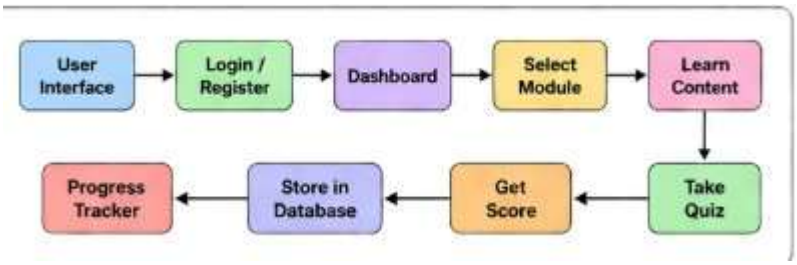
3. AI Content Generation Technique

- Uses AI prompts to generate explanations, examples, and visuals
- Converts complex environmental topics into simple learning content

4. Progress Tracking Algorithm

- Updates completion status after each module
- Stores user performance data in database.

3.2.2 Block Diagram



3.2.3 Hardware and Software Specifications

The Eco Wonder application is designed to run on standard computing devices and mobile systems with internet connectivity.

Table 3.2 Hardware details

Processor	2 GHz Intel or higher
Storage	500 MB free space
RAM	4 GB minimum (8 GB recommended)

Table 3.3 Software details

Operating System	Windows 10 / Android / iOS
Programming Language	JavaScript / Dart / Python (AI support)
Database	Firebase / Supabase

Chapter 4

Implementation

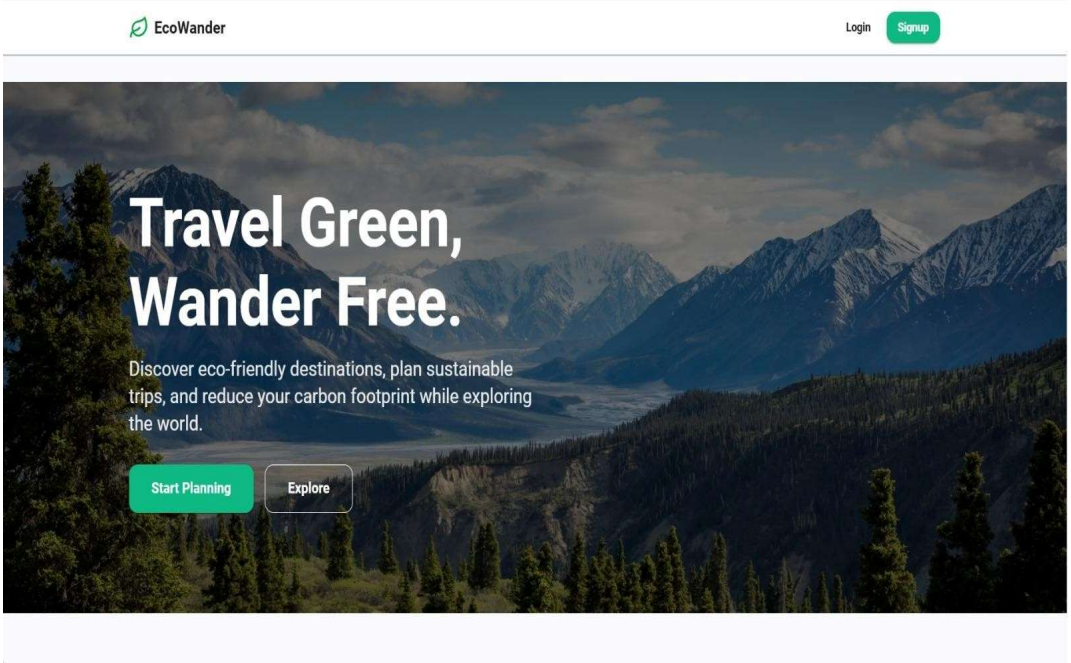
4.1 System Overview

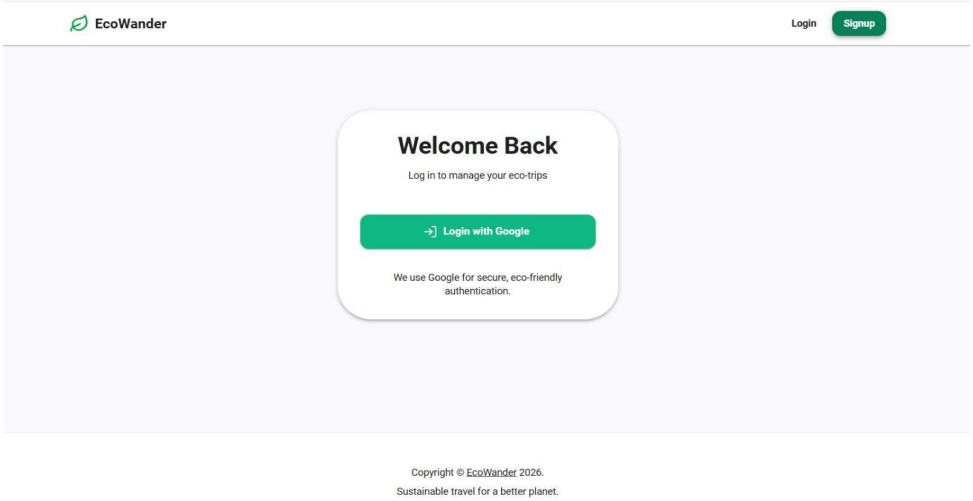
The Eco Wonder application is developed as an AI-based system that promotes environmental awareness by providing eco-friendly suggestions and solutions. The system consists of a user-friendly interface where users can input their queries or activities related to sustainability. The input is processed using an AI-based system that analyzes the data and generates relevant eco-friendly recommendations.

The application follows a simple architecture where the user interacts with the mobile or web interface, which is connected to the backend system. The backend processes the request using AI algorithms and retrieves or generates appropriate responses. These responses are then displayed to the user in an easy-to-understand format. The system also stores user data and interactions in a database for future reference and improvement of recommendations.

4.2 Project working with Snapshots

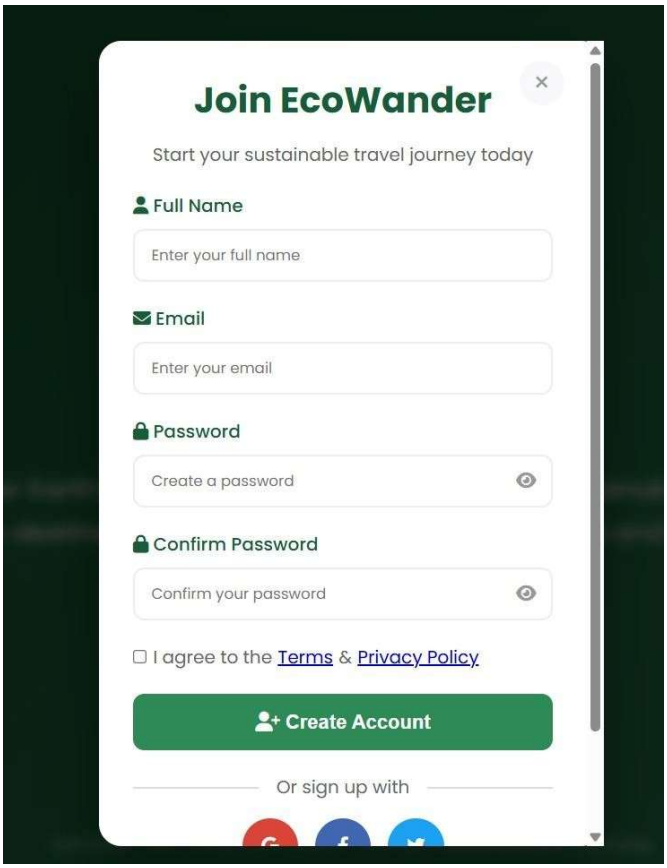
The working of the Eco Wonder application can be explained in the following steps:





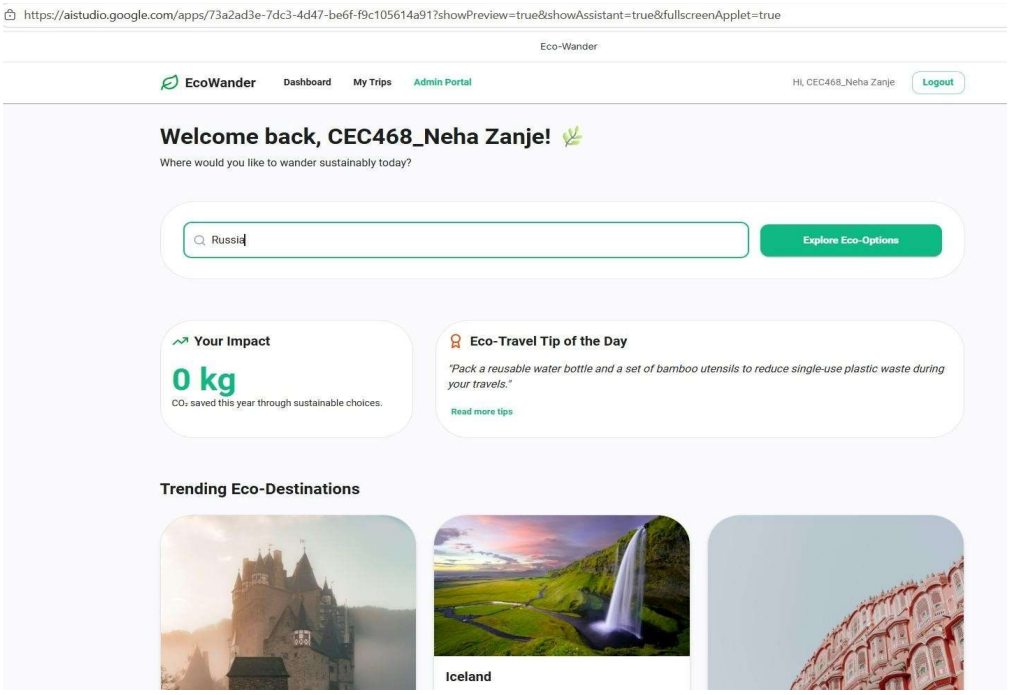
4.2.1 User Input:

The user enters a query or selects an eco-related activity .



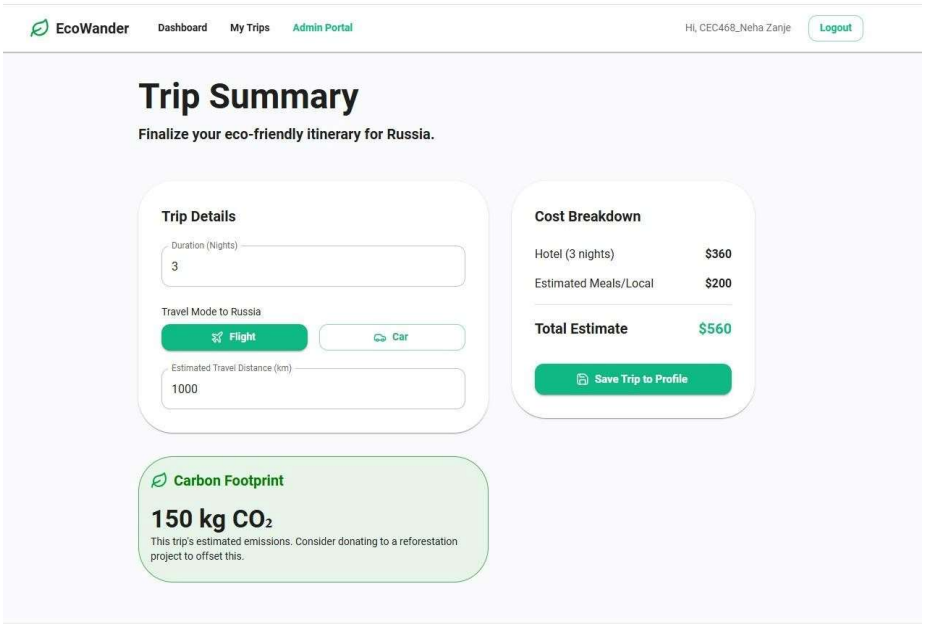
4.2.2 Processing:

The system processes the input using AI algorithms.



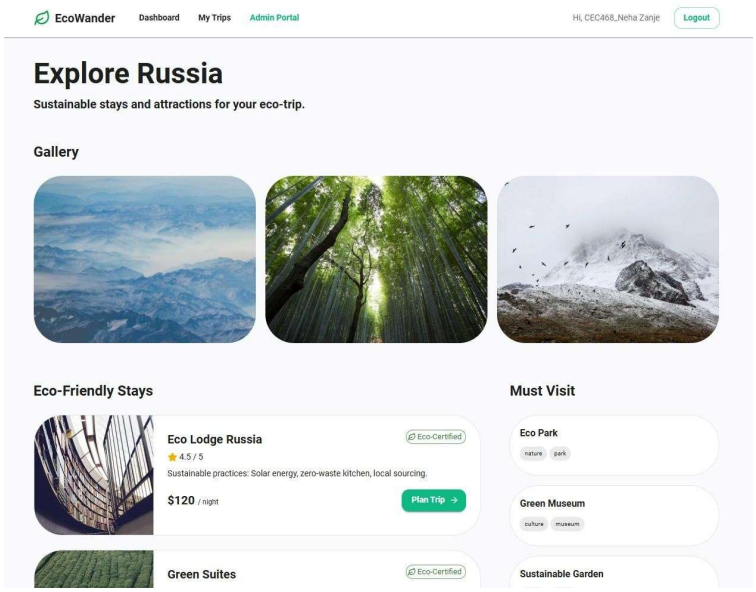
4.2.3 AI Response Generation:

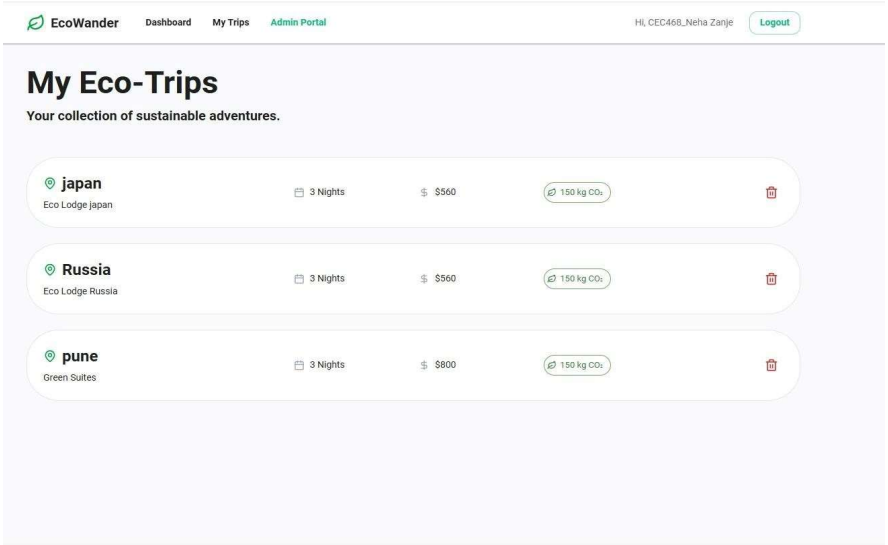
The AI generates eco-friendly suggestions or solutions.



4.2.4 Display Output:

The results are displayed on the screen.





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Chapter 5

Applications and Future Scope

There are various applications of this domain system. The application is listed here.

5.1 Applications

The Eco Wonder application has several practical applications in real life. It can be used to promote environmental awareness among students and the general public. The application helps users learn simple and effective ways to adopt eco-friendly habits such as saving water, reducing waste, and conserving energy.

It can also be used in educational institutions as a learning tool to teach sustainability concepts. Organizations and communities can use the application to encourage green practices and spread awareness about climate change. Additionally, it can support campaigns and initiatives aimed at protecting the environment.

5.2 Future Scope

The future scope of the Eco Wonder application is wide and promising. The application can be enhanced by integrating advanced AI models to provide more accurate and personalized suggestions. Features like voice input, multilingual support, and real-time environmental data can be added to improve user experience.

In the future, the app can also include features such as tracking carbon footprint, gamification (reward system), and community challenges to motivate users. Integration with IoT devices and smart home systems can further help users monitor and reduce their environmental impact. Overall, the application has the potential to grow into a powerful tool for promoting sustainable living globally.

Chapter 6

Summary

The Eco Wonder application is an AI-based system developed to promote environmental awareness and sustainable living. It allows users to input queries related to eco-friendly practices and provides intelligent suggestions using artificial intelligence. The system includes a user interface, AI processing unit, and a database to store and manage user data efficiently.

The application is useful in spreading awareness about environmental conservation and encouraging users to adopt green habits in their daily lives. With future improvements like advanced AI features, real-time data, and interactive tools, Eco Wonder has the potential to become a powerful platform for supporting a sustainable and eco-friendly future.

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